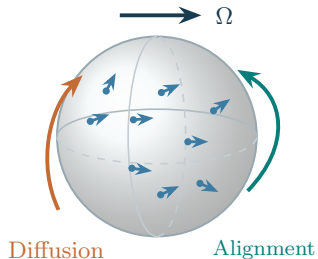


I Frozen-field relaxation

Rotational noise and alignment



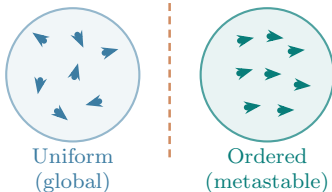
Fixed J : $r = k(J)$
Angular rate: $\tau_{\text{noise}} \lambda_d(r) > 0$

II Feedback and basins

One density, two initial states

$$J \rightarrow r = k(J) \rightarrow \rho c_d(r) = J$$

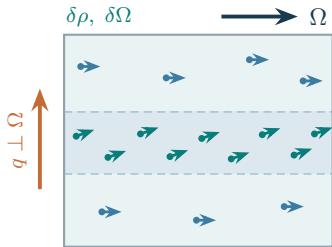
Basin boundary (schematic)



Homogeneous PDE, $\rho = 1.90$

III Transverse response

Coupled density and orientation



$$\Theta < 0 \implies v_{\pm} \notin \mathbb{R}$$

Transverse first-order SOH
 $K_2 > 0$: diffusive dispersion